

Day 1 Project Setup & Patient Management Module

Date

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Overview

Today marks the foundation of the **Clinic Management System**. The goal was to initialize the Spring Boot project, establish MongoDB connectivity, implement the first core module (Patient Management), and create both REST API endpoints and a Thymeleaf-based web interface.

Technology Stack Used

Technology	Version	Purpose
Java	21 (compiled on JDK 26)	Programming Language
Spring Boot	3.4.1	Backend Framework
Spring Data MongoDB	3.4.1	Database Integration
MongoDB	Latest (local)	NoSQL Database
Thymeleaf	3.4.1	Server-side HTML templating
Bootstrap	5.3.2 (CDN)	UI Styling
Spring Validation	3.4.1	Request DTO validation
SpringDoc OpenAPI	2.6.0	API Documentation (prepared)
Maven	Bundled with IntelliJ	Build & Dependency Management
Git	Latest	Version Control

What Was Built Today

1. Project Initialization

- Created Maven-based Spring Boot project with `pom.xml`

- Configured application.yml with MongoDB connection string (mongodb://localhost:27017/clinic_db)
- Set server port to 8080
- Enabled MongoDB auditing (@EnableMongoAuditing) for automatic createdAt and updatedAt timestamps

2. Package Structure Established

Created the complete layered architecture as per project requirements:

```
com.clinic/
├── ClinicManagementSystemApplication.java
├── config/                (reserved for future: SecurityConfig, OpenApiC
onfig)
├── controller/           (REST + Thymeleaf controllers)
├── dto/                  (Request/Response Data Transfer Objects)
├── entity/               (MongoDB Document entities)
├── enums/                (Java enums for type safety)
├── exception/            (reserved for future: custom exceptions)
├── repository/           (Spring Data MongoDB repositories)
├── security/             (reserved for future: JWT configuration)
├── service/              (Business logic layer)
├── mapper/               (reserved for future: entity-DTO mappers)
└── util/                 (reserved for future: utility classes)
```

3. Enums Created

Two enums were defined to ensure data consistency and avoid string literals in the database:

Gender.java - MALE - FEMALE - OTHER

BloodGroup.java - A_POSITIVE, A_NEGATIVE - B_POSITIVE, B_NEGATIVE - AB_POSITIVE, AB_NEGATIVE - O_POSITIVE, O_NEGATIVE

4. Patient Entity (Patient.java)

Created the core Patient document mapped to MongoDB collection patients:

Field	Type	Description
patientId	String	Custom ID (e.g., PAT-A1B2C3D4)
name	String	Patient full name
email	String	Contact email
phone	String	Contact phone

Field	Type	Description
gender	Gender	Enum: MALE/FEMALE/OTHER
dateOfBirth	LocalDate	Date of birth
address	String	Residential address
bloodGroup	BloodGroup	Enum: A_POSITIVE, O_NEGATIVE, etc.
createdAt	LocalDateTime	Auto-populated by @CreatedDate
updatedAt	LocalDateTime	Auto-populated by @LastModifiedDate

Annotations used: @Document, @Id, @CreatedDate, @LastModifiedDate

5. DTOs (Data Transfer Objects)

PatientRequestDto.java Used for incoming API requests: - Validation annotations applied: - @NotBlank ON name, email, phone - @Email ON email - @Size(min=10, max=15) ON phone - @NotNull ON gender, dateOfBirth, bloodGroup

PatientResponseDto.java Used for outgoing API responses: - Mirrors entity fields including patientId, createdAt, updatedAt - Separates internal entity from external contract

6. Repository Layer (PatientRepository.java)

Extends MongoRepository<Patient, String> with custom query methods: - findByEmail(String email) For duplicate checking (future use) - findByPhone(String phone) For duplicate checking (future use) - findByNameContainingIgnoreCase(String name) For search functionality

7. Service Layer (PatientService.java + PatientServiceImpl.java)

Interface: Defines the contract for patient operations.

Implementation: Contains business logic: - createPatient() Generates custom PAT-XXXX ID using UUID, saves to MongoDB - getAllPatients() Retrieves all patient documents - getPatientById() Finds by patientId, throws RuntimeException if not found (will be replaced with custom exception on Day 2) - updatePatient() Full update of all fields - deletePatient() Deletes by ID after verifying existence - searchPatientsByName() Case-insensitive name

search - mapToResponseDto() Private helper method to convert Patient entity → PatientResponseDto

Design pattern: Constructor-based dependency injection (no @Autowired on fields).

8. REST API Controller (PatientRestController.java)

Base path: /api/patients

Method	Endpoint	Description	Status Codes
POST	/api/patients	Create new patient	201 Created
GET	/api/patients	List all patients	200 OK
GET	/api/patients/{id}	Get patient by ID	200 OK
PUT	/api/patients/{id}	Update patient	200 OK
DELETE	/api/patients/{id}	Delete patient	204 No Content
GET	/api/patients/search?name=	Search by name	200 OK

Features: - @Valid annotation triggers DTO validation - ResponseEntity used for proper HTTP status codes - @RequestBody for JSON input, @PathVariable for IDs, @RequestParam for search

9. Thymeleaf Web Controllers

IndexController.java Serves the homepage at /

PatientViewController.java Serves patient list at /patients - Injects PatientService to fetch all patients - Adds patients attribute to Spring Model - Returns view name patients/list

10. Thymeleaf HTML Templates

templates/index.html Landing page - Bootstrap 5 navbar with navigation links - Welcome message and link to patient list - Responsive container layout

templates/patients/list.html Patient listing page - Bootstrap 5 styled table with dark header - Thymeleaf th:each iteration over patient list - Displays all 9 patient fields in columns - Empty state message when no patients exist - Badge showing total patient count - Alert box listing all available REST endpoints for reference

Bootstrap integration: Via CDN (no local CSS files yet): - CSS:

<https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css>

- JS:

<https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js>

Testing Performed

REST API Tests (via Postman / Browser)

1. Create Patient

POST <http://localhost:8080/api/patients>

```
{
  "name": "Ali Khan",
  "email": "ali@example.com",
  "phone": "03001234567",
  "gender": "MALE",
  "dateOfBirth": "1990-05-15",
  "address": "123 Main Street, Karachi",
  "bloodGroup": "O_POSITIVE"
}
```

- Result: 201 Created with auto-generated patientId: "PAT-A1B2C3D4"
- createdAt and updatedAt auto-populated

2. List Patients

- GET <http://localhost:8080/api/patients>
- Result: Array containing the created patient

3. Get By ID

- GET <http://localhost:8080/api/patients/PAT-A1B2C3D4>
- Result: Single patient object

4. Update Patient

- PUT <http://localhost:8080/api/patients/PAT-A1B2C3D4>
- Result: Updated patient with new updatedAt timestamp

5. Search

- GET <http://localhost:8080/api/patients/search?name=Ali>

- Result: Filtered list matching “Ali” (case-insensitive)

6. Delete

- DELETE `http://localhost:8080/api/patients/PAT-A1B2C3D4`
- Result: 204 No Content

Thymeleaf UI Tests

- `http://localhost:8080/` → Homepage renders correctly
- `http://localhost:8080/patients` → Table renders with patient data after REST creation
- Empty state displays correctly when database is clean

Database Verification

- Opened **MongoDB Compass**
- Connected to `mongodb://localhost:27017`
- Database `clinic_db` created automatically
- Collection `patients` contains documents with custom `patientId` as `_id`
- `createdAt` and `updatedAt` fields present as `ISODate`

Challenges Faced & Solutions

Challenge	Solution
Java 26 compiler incompatibility with Lombok	Removed Lombok entirely; switched to manual getters/setters/constructors
Java 26 <code>TypeTag :: UNKNOWN</code> compiler crash	Downgraded project compilation target from Java 26 to Java 21 while keeping JDK 26 installed
<code>mvn</code> command not recognized in PowerShell	Used IntelliJ built-in Maven panel (Lifecycle → clean → install) instead of terminal
ThemeSource <code>NoClassDefFoundError</code> on	Cleaned Maven cache and reloaded

Challenge	Solution
startup	dependencies
Git push failed (src refspec main does not match any)	Committed locally first (git commit) before pushing to remote

Git Activity

- **Repository:** <https://github.com/abdul15x/clinic-management-system>
- **Branch:** main
- **Commit:** feat: project setup and patient basic CRUD
- **Files pushed:** All source code, pom.xml, application.yml, HTML templates

Files Created Today (Complete List)

```
pom.xml
src/main/resources/application.yml
src/main/resources/templates/index.html
src/main/resources/templates/patients/list.html
src/main/java/com/clinic/ClinicManagementSystemApplication.java
src/main/java/com/clinic/enums/Gender.java
src/main/java/com/clinic/enums/BloodGroup.java
src/main/java/com/clinic/entity/Patient.java
src/main/java/com/clinic/dto/PatientRequestDto.java
src/main/java/com/clinic/dto/PatientResponseDto.java
src/main/java/com/clinic/repository/PatientRepository.java
src/main/java/com/clinic/service/PatientService.java
src/main/java/com/clinic/service/PatientServiceImpl.java
src/main/java/com/clinic/controller/PatientRestController.java
src/main/java/com/clinic/controller/PatientViewController.java
src/main/java/com/clinic/controller/IndexController.java
```

What's Next (Day 2 Preview)

- Add custom exceptions (PatientNotFoundException, DuplicateEmailException)
- Replace RuntimeException with proper exception handling
- Implement pagination: GET /api/patients?page=0&size=10

- Add Thymeleaf patient creation form (`patients/form.html`)
 - Add Thymeleaf patient detail view (`patients/detail.html`)
 - Add `@RestControllerAdvice` global exception handler (Day 11 target, but basic setup on Day 2)
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How to Run This Version

1. Ensure MongoDB is running on `localhost:27017`
2. Open project in IntelliJ IDEA
3. Set Project SDK to your installed JDK (Java 26 is fine)
4. Set Language Level to 21
5. Run `ClinicManagementSystemApplication.java` (green play button)
6. Access:
 - Web UI: `http://localhost:8080/`
 - Patient List: `http://localhost:8080/patients`
 - REST API: `http://localhost:8080/api/patients`